

Review retrospective:
*generic even-monomial Gibbs moments, a scope note overtaken
by its own file*

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Abstract

A soundness review of `Laplace/OneD/MonomialPotential.lean`: the generic pure even-monomial Gibbs moments for $L_k(x) = x^{2k}/(2k)!$ — the closed forms $\int_{\mathbb{R}} x^{2j} e^{-tL_k} = (1/k)((2k)!/t)^{(2j+1)/(2k)} \Gamma((2j+1)/(2k))$, the partition function, the even/odd expectations, and the polynomial-times-Gibbs integrability. Result: **a clean fidelity pass with three landed edits** — a stale scope-note docstring and two redundant imports, the imports cleared by a full project build.

1 Setting

This is the generic- k foundation of the monomial / J-function family: the quartic ($k = 2$), sextic ($k = 3$), and the `TwoD/KthKth*` separable-potential asymptotics all specialise from it. Six files import it, so it is a genuine foundation, reviewed (like Anharmonic) to validate the base its consumers stand on.

2 What was reviewed

Eleven declarations: the potential and its `simp` lemma; a private bound $x^2 \leq 1 + x^{2k}$; two integrability theorems; the half-line and full-line moment closed forms; the odd-moment vanishing; the partition function and its positivity; and the even/odd Gibbs expectations.

3 Fidelity / soundness findings

Sound; every closed form checks out. The half-line constant $1/(2k)$ doubles (via `integral_comp_abs`) to the full-line $1/k$; the expectation exponent simplifies as $(2j+1)/(2k) - 1/(2k) = 2j/(2k) = j/k$; the Gamma arguments and the $((2k)!/t)$ bases are right; and the integrability rests on the Gaussian domination $x^2 \leq 1 + x^{2k}$, which — importantly — holds *uniformly for all* $k \geq 1$ ($x^2 \leq 1$ on the unit interval; $x^{2k} = (x^2)^k \geq x^2$ outside it). GPT re-derived each and found no mismatch.

Two benign notes, recorded but not escalated: `kth_moment_odd` and `gibbsExpectation_kthPotential_odd` omit the $k \geq 1$ and $t > 0$ hypotheses. This is a correct *generalisation*, not a defect: odd-moment vanishing follows from reflection symmetry alone (plus Mathlib's convention that the integral of a non-integrable function is 0), so it holds for every

k, t . The statement is cleaner than the Gibbs-regime motivation, and adding the hypotheses would only narrow it (and change the signature).

4 Simplifications applied

Three landed.

- **A stale scope note.** The docstring’s “Scope note (deferred)” announced that the integrability witnesses `kth_integrable_pow/_pot` were “deferred to a follow-up tide” and warned that the generic Gaussian domination “needs case analysis on $k = 1$ vs $k \geq 2$ ”. Both claims are contradicted by the file itself: the lemmas are proved right there, and *uniformly* for $k \geq 1$ via $x^2 \leq 1 + x^{2k}$ — the feared case split never materialised. Rewritten to describe what the file actually does. (A comment-only fix.)
- **Two redundant imports.** `Gamma.Basic` (the `Real.Gamma` API arrives via `GaussianIntegral/Integral.Gamma`) and `Haar.NormedSpace` (its `integral_comp_mul_right` is unused here; `integral_comp_abs` comes from `Lebesgue.Integral`), both removed.

Because this is a foundation file, the import removals were arbitrated by a **full project build**, not a single-module one: `lake build` went green at 8306 jobs (unchanged from baseline) with all six consumers rebuilding. The reviewer had flagged `Haar.NormedSpace` as the weaker candidate — it might have supplied `integral_comp_abs/integral_neg_eq_self` transitively — and the full build is precisely what settled that doubt: the lemmas still resolve, so the import was redundant. Eleven declarations byte-identical; GPT affirmed all three changes, noting (correctly) that the whole-project rebuild, not the unchanged job count alone, is the load-bearing evidence.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A “deferred” scope note is worth checking against the file it sits in.** Documentation written while a result was still future can survive into the commit that delivers it. Here the note even preserved a difficulty (the $k = 1$ vs $k \geq 2$ split) that the eventual uniform proof sidestepped — so the note mis-described both the *status* and the *shape* of the work. Grepping the file for the very lemma names the note calls “deferred” is the one-line check.
- **The full build dissolves a reviewer’s transitive-import doubt.** When the reviewer cannot tell from text whether an import is load-bearing transitively, do not argue it — remove it and rebuild the whole project. A green whole-project build is a stronger and simpler answer than tracing the dependency graph by hand, and it is the right tool whenever the file has consumers.

7 Follow-ups

- The generic-monomial base is now reviewed; its specialisations (`Quartic`, `Sextic`) and the `TwoD/KthKth*` lifts are natural next targets, as are the J-function files. The `Multi/Covariance*` trio remains the large sectioned-approach target.